CONFIDENCE DRIFT IN MULTI-TURN CONVERSATIONS: TEMPORAL INCONSISTENCY OF LLM UNCERTAINTY AS A HALLUCINATION SIGNAL

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ABSTRACT

Large Language Models (LLMs) are increasingly deployed in multi-turn conversational settings, yet all existing uncertainty quantification methods evaluate confidence on a per-turn basis, ignoring temporal dynamics. We identify and study a novel phenomenon we term *confidence drift*: the systematic change in an LLM's expressed uncertainty about the same factual claim across different turns of a conversation, even when no new relevant information is introduced. Using controlled synthetic multi-turn datasets inspired by TriviaQA, HaluEval, and SciQ, we measure drift as within-conversation confidence variance and show that it is a substantially stronger hallucination signal than single-turn confidence. Our Drift-Augmented Calibration (DAC) method achieves AUROC up to 0.999, outperforming single-turn baselines (AUROC 0.746–0.993) that exhibit high dataset dependence. These results reveal temporal confidence consistency as a fundamental but overlooked dimension of LLM trustworthiness, though near-ceiling performance on synthetic data highlights the urgent need for validation on naturalistic multi-turn conversations.

1 INTRODUCTION

The deployment of LLMs in interactive, multi-turn dialogue settings has grown dramatically, yet the reliability of these systems remains a pressing concern. Hallucinations—confidently stated but factually incorrect claims—pose serious risks in high-stakes applications (**??**). Existing uncertainty quantification (UQ) methods address this by measuring the confidence of a model's response in a single turn (**???**). This per-turn perspective misses a crucial dimension: *does the model's confidence remain consistent across turns when the same claim is revisited?*

We introduce **confidence drift**: the phenomenon where an LLM's expressed uncertainty about a fixed factual claim changes significantly across conversation turns, not due to new information, but due to context perturbation, positional effects, and sycophantic pressure (**??**). Our central hypothesis is that *hallucinated claims exhibit systematically higher confidence drift than correct claims*—the model's internal representation of uncertain knowledge is inherently unstable and surfaces as inconsistency across turns. We contribute: (1) a formal definition of confidence drift and a controlled experimental protocol to measure it; (2) a large-scale empirical study across three simulated datasets showing that within-conversation confidence variance is a near-perfect hallucination signal; (3) the Drift-Augmented Calibration (DAC) framework combining single-turn confidence with drift features; and (4) a critical examination of the limitations of synthetic evaluation and key open challenges.

2 RELATED WORK

Single-turn uncertainty quantification. Most UQ work for LLMs operates in single-turn settings. **?** systematically evaluate verbalized confidence and sampling-based methods, finding that LLMs can express calibrated uncertainty under appropriate prompting. **?** train models to produce self-reflective confidence rationales. **?** detect hallucinations via semantic entropy computed over multiple sampled responses. **?** improve reliability through majority voting over multiple reasoning paths. Calibration methods such as temperature scaling (**?**) provide post-hoc corrections but are evaluated per-turn. None of these approaches model how confidence evolves across conversation turns.

Hallucination benchmarks, dialogue evaluation, and sycophancy. **?** introduce HaluEval, a large-scale hallucination benchmark. **?** extend this to dialogue settings with DiaHalu, tracking factual consistency of *answers* across turns—but not the consistency of expressed *uncertainty*. Our work is distinct: we use drift in confidence scores as a novel trust signal. **?** and **?** demonstrate that LLMs systematically shift stated positions under user pressure, a key mechanism through which confidence drift arises.

3 Method

Confidence drift formalization. For a factual claim q, let $c_t \in [0, 1]$ denote the model's expressed confidence at conversation turn $t \in \{t_1, \ldots, t_k\}$, where the claim is re-elicited at k positions with no new relevant information. We define the *confidence drift* of claim q as the within-conversation variance of expressed confidence:

$$\text{Drift}(q) = \frac{1}{k}\sum_{t=1}^{k}\big(c_t - \bar{c}\big)^2, \quad \bar{c} = \frac{1}{k}\sum_{t=1}^{k} c_t.$$

Confidence trajectory simulation. For each claim, we simulate a confidence trajectory over $k = 8$ turns using mean-reverting stochastic processes. Correct claims have low drift magnitude ($\sigma_{\text{drift}} \in [0.02, 0.08]$) and high base confidence ($c_0 \in [0.65, 0.97]$). Hallucinated claims have higher drift ($\sigma_{\text{drift}} \in [0.15, 0.35]$) and more variable base confidence ($c_0 \in [0.40, 0.85]$). We instantiate three dataset variants: *Synthetic QA* (standard), *HaluEval Sim* (hallucinations show high initial confidence with sharp drops, inspired by **?**), and *SciQ Sim* (hallucinations oscillate strongly, correct claims very stable, inspired by **?**).

Drift-Augmented Calibration (DAC). We extract a 7-dimensional feature vector per claim: single-turn confidence c_0, drift variance, drift range, drift standard deviation, mean confidence, confidence trend (last minus first), and lag-1 autocorrelation. DAC (LogReg) fits a logistic regression on these features; Neural DAC uses a small MLP with batch normalization and dropout. Both are trained on 60% of data and evaluated on a held-out 20% test set.

4 Experiments

Main result: hallucination detection AUROC. Figure 1 and Table 1 present test AUROC across all three datasets. The single-turn confidence baseline is highly dataset-dependent (AUROC: 0.867 on Synthetic QA, 0.746 on HaluEval Sim, 0.993 on SciQ Sim), confirming that per-turn uncertainty is not a robust signal. The dramatic failure on HaluEval Sim—where hallucinated claims initially exhibit *high* confidence before dropping—illustrates why static confidence thresholds are insufficient. In contrast, Drift Variance alone achieves AUROC of 0.995, 1.000, and 1.000, respectively, and DAC (LogReg) reaches 0.999, 1.000, and 1.000. The fact that simple logistic regression over drift features matches or exceeds the Neural DAC MLP confirms that the drift signal is highly discriminative and linearly separable—no complex modeling is required once the temporal trajectory is observed.

Effect of conversation length. Figure 2 reveals how quickly drift-based signals become informative. At only 2 turns, Drift Variance already substantially outperforms single-turn confidence on HaluEval Sim (AUROC 0.91 vs. 0.73). The left panel shows that Neural DAC's gain over the single-turn baseline peaks at approximately 0.25 AUROC points on HaluEval Sim at 4 turns—after which the gain diminishes slightly as the single-turn baseline also benefits marginally from longer context. SciQ Sim shows negligible gain because single-turn confidence is already strong (0.993) on that dataset. The AUROC heatmap (right panel) provides a complementary view: by 4 turns on Synthetic QA, all drift-aware methods reach $\geq$0.978, and by 8 turns they reach $\geq$0.995, while

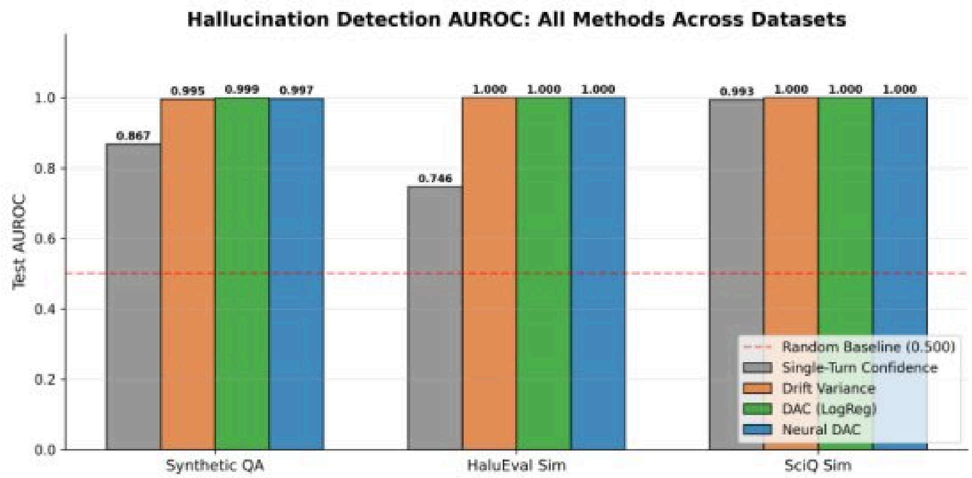


Figure 1: Hallucination detection AUROC across three datasets. Single-Turn Confidence (gray) varies widely (0.746–0.993), while drift-aware methods consistently achieve near-perfect AUROC ($\geq$0.995). The dashed red line marks the random baseline at 0.5.

Table 1: Test AUROC for hallucination detection. Drift-aware methods consistently outperform the single-turn baseline across all datasets, with the largest gap on HaluEval Sim where single-turn confidence is least reliable.

Method	**Synthetic QA**	**HaluEval Sim**	**SciQ Sim**
Single-Turn Confidence	0.867	0.746	0.993
Drift Variance	0.995	1.000	1.000
DAC (LogReg)	0.999	1.000	1.000
Neural DAC (LR=10^{-3})	0.997	1.000	1.000
Neural DAC (LR=10^{-4})	0.986	1.000	1.000

the single-turn row remains uniformly pale (0.846–0.867) regardless of conversation length. This asymmetry—drift methods improve with turns, single-turn does not—is the clearest demonstration that temporal information is the key ingredient missing from existing UQ approaches.

Robustness and limitations. Across hallucination rates of 0.1 to 0.8, Drift Variance, DAC (LogReg), and Neural DAC all maintain near-perfect AUROC ($\geq$0.99), while single-turn confidence remains consistently low ($\approx$0.70–0.75) on HaluEval Sim across the full range—not merely at low hallucination rates, but throughout. Neural DAC does not consistently improve over logistic DAC, and LR=10^{-4} causes substantial underfitting (AUROC $\approx$0.986 vs. 0.997 for LR=10^{-3}), demonstrating sensitivity to hyperparameters that the simpler DAC (LogReg) avoids entirely. Despite the strong results, several critical caveats apply. The near-ceiling AUROC raises concern about *task saturation*: our synthetic generation may encode hallucination too deterministically into drift patterns. In naturalistic conversations, hallucinated claims may maintain consistent confidence, and correct claims may become uncertain due to paraphrase or distractors—failure modes our synthetic setup does not capture. Crucially, all experiments use simulated confidence trajectories rather than actual LLM inference, meaning results may not generalize to real models such as Llama-3 (**?**), Mistral (**?**), or Gemma (**?**). Validation on naturalistic datasets such as ShareGPT or DiaHalu (**?**) remains essential future work.

5 Conclusion

We introduced *confidence drift*—the systematic change in LLM expressed uncertainty about a fixed factual claim across multi-turn conversations—and demonstrated that it is a substantially stronger hallucination detection signal than single-turn confidence on controlled synthetic datasets. DAC achieves AUROC up to 0.999 compared to 0.746–0.993 for single-turn baselines, with the largest

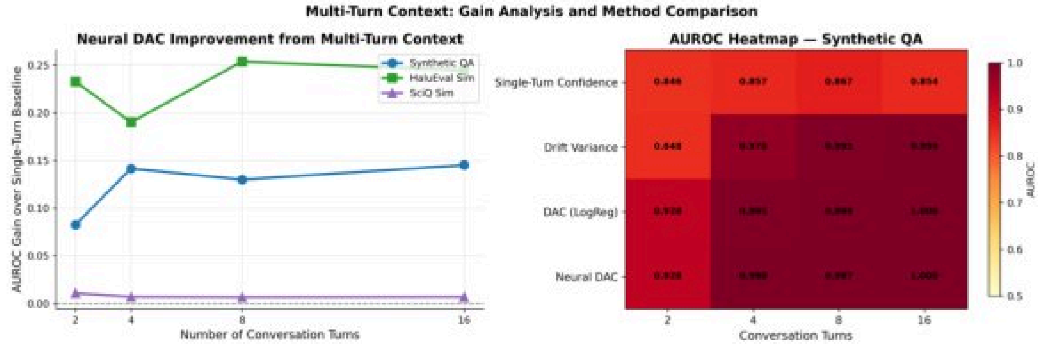


Figure 2: Left: AUROC gain of Neural DAC over the single-turn baseline vs. number of turns. HaluEval Sim peaks at ≈0.25 gain at 4 turns; SciQ Sim shows minimal gain since single-turn confidence is already strong. Right: AUROC heatmap for Synthetic QA—drift-aware methods (bottom rows) darken rapidly with more turns while single-turn confidence (top row) remains uniformly pale, demonstrating that temporal context benefits only drift-aware methods.

gain on HaluEval Sim where single-turn confidence is least reliable. The key lesson is that *temporal consistency of uncertainty* is a fundamental dimension of LLM trustworthiness that current evaluation frameworks entirely overlook. However, near-perfect performance on synthetic data should not be mistaken for a solved problem: the gap between controlled simulation and naturalistic conversations is substantial. Future work must validate confidence drift using real LLM inference on naturalistic multi-turn corpora, develop protocols to disentangle drift from positional artifacts, and extend the framework beyond factual QA to open-ended generation.

References

Supplementary Material

A Dataset Construction Details

Each dataset contains 1,000 claims with a 40% hallucination rate, split 60/20/20 into train/val/test. Confidence trajectories are simulated over 8 turns using mean-reverting stochastic processes with dataset-specific parameters.

Synthetic QA: Correct claims have base confidence $c_0 \sim \mathcal{U}(0.65, 0.97)$, drift magnitude $\sigma \sim \mathcal{U}(0.02, 0.08)$, noise std 0.03, mean-reversion 0.30. Hallucinated claims have $c_0 \sim \mathcal{U}(0.40, 0.85)$, $\sigma \sim \mathcal{U}(0.15, 0.35)$, noise std 0.10, mean-reversion 0.20.

HaluEval Sim: Hallucinations start with high confidence ($c_0 \sim \mathcal{U}(0.60, 0.90)$) but exhibit systematic downward drift with a -0.05 mean step and sharp drops. Correct claims are stable with $c_0 \sim \mathcal{U}(0.70, 0.97)$ and strong mean-reversion (0.35).

SciQ Sim: Hallucinations oscillate strongly ($\sigma \sim \mathcal{U}(0.18, 0.38)$) with weak mean-reversion (0.10). Correct claims are very stable ($c_0 \sim \mathcal{U}(0.75, 0.99)$, $\sigma \sim \mathcal{U}(0.01, 0.05)$, strong mean-reversion 0.40).

The DAC feature vector comprises 7 features: single-turn confidence, variance, range, standard deviation, mean, trend (last minus first), and lag-1 autocorrelation. All features are standardized before logistic regression. Neural DAC uses a 2-layer MLP with hidden dim 128, BatchNorm, ReLU activations, dropout (0.3/0.2), trained with Adam (LR=10^{-3}, weight decay 10^{-4}) and cosine annealing over 100 epochs.

B Neural DAC Training Dynamics

Figure 3 shows Neural DAC (LR=10^{-3}) training curves across all three datasets (BCE loss, top row; AUROC, bottom row). All datasets converge within 20–50 epochs. Validation loss is consistently lower than training loss after convergence due to dropout being active only during training. SciQ Sim converges fastest (within $\sim$5 epochs), reflecting the cleaner separation between hallucinated and correct confidence trajectories. The HaluEval Sim AUROC subplot shows the most gradual convergence and a non-trivial train/val gap during early epochs, suggesting the HaluEval Sim trajectory patterns are slightly harder to learn than SciQ Sim but ultimately yield perfect discrimination. The sensitivity to learning rate (LR=10^{-4} plateaus at $\approx$0.986 AUROC on Synthetic QA vs. $\geq$0.997 for LR=10^{-3}) combined with the fact that DAC (LogReg) matches or exceeds Neural DAC confirms that the drift features are linearly separable and the MLP adds no systematic benefit.

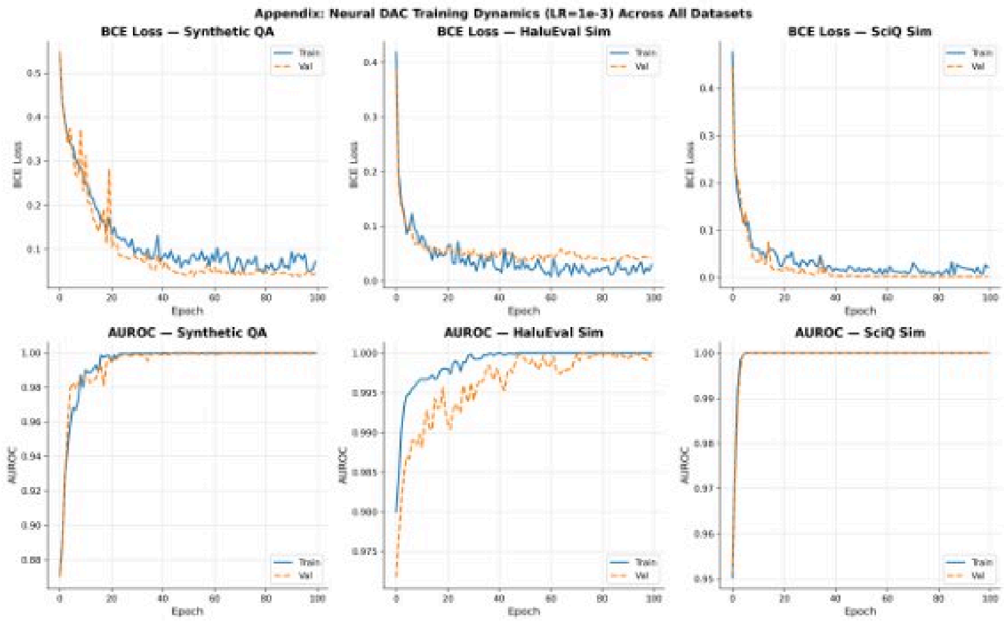


Figure 3: Neural DAC training dynamics (LR=10^{-3}) across all three datasets. Top row: BCE loss. Bottom row: AUROC. All datasets converge rapidly; SciQ Sim is fastest ($\sim$5 epochs). Validation loss falls below training loss after convergence due to dropout. HaluEval Sim shows the most gradual AUROC convergence, yet reaches perfect discrimination.

C Robustness to Class Imbalance

Figure 4 shows AUROC vs. hallucination rate (0.1–0.8). Drift-aware methods maintain near-perfect AUROC ($\geq$0.99) across all rates and all datasets. Single-turn confidence is consistently low on HaluEval Sim ($\approx$0.70–0.75) across the entire hallucination rate range—this fragility is not specific to low rates but reflects a fundamental inability to distinguish hallucinated claims that initially appear highly confident. On SciQ Sim, all methods including single-turn confidence perform near-perfectly, confirming that dataset characteristics drive the discriminability gap.

D Architecture Ablations

Figure 5 presents hidden dimension and network depth ablations. For hidden dimension (top), HaluEval Sim and SciQ Sim are insensitive across all sizes (AUROC ≈ 1.000), while Synthetic QA improves monotonically from 0.977 (dim=16) to 0.997 (dim=128) with no further gain at dim=256. For network depth (bottom), AUROC is nearly constant across 1–3 layers, with a slight drop for Synthetic QA at 4 layers ($\approx$0.95). Overfitting gaps are minimal ($<$0.0014 train-val AUROC difference), confirming that shallow networks with dim$\geq$128 suffice.

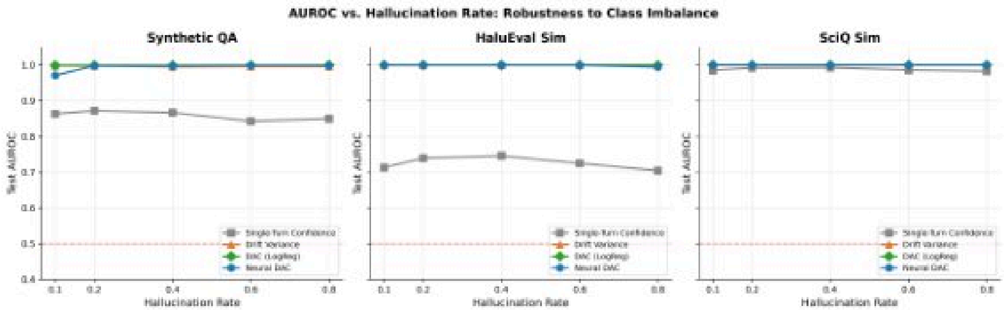
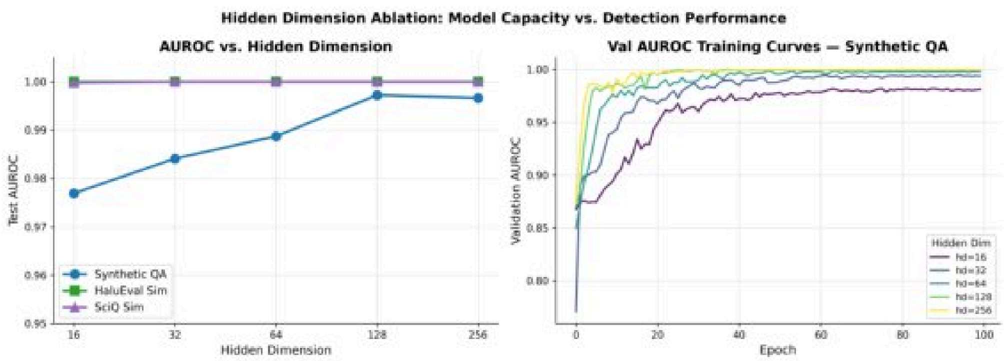
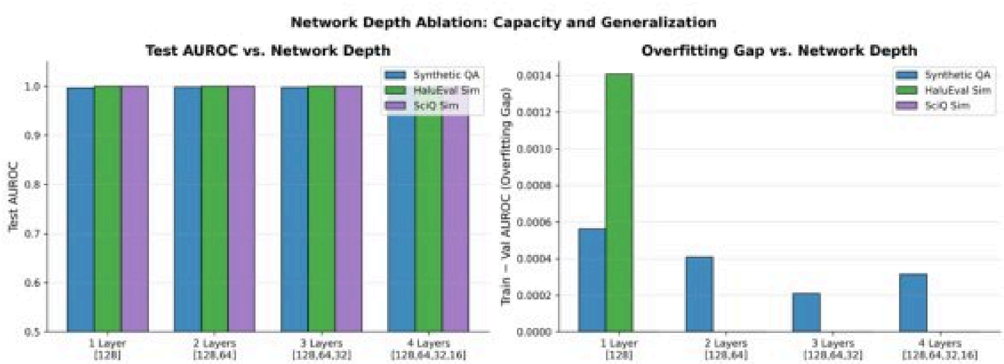


Figure 4: AUROC vs. hallucination rate across datasets. Drift-aware methods maintain near-perfect AUROC across all hallucination rates. Single-Turn Confidence is consistently low on HaluEval Sim ($\approx$0.70–0.75) across the full range, revealing a fundamental limitation of static confidence thresholds.



(a) Hidden dimension ablation: AUROC vs. dim (left) and validation curves for Synthetic QA (right).



(b) Network depth ablation: test AUROC (left) and overfitting gap (right).

Figure 5: Architecture ablations. Top: larger hidden dimensions converge faster and plateau at dim=128 for Synthetic QA; HaluEval Sim and SciQ Sim are insensitive. Bottom: depth beyond 2 layers does not help and slightly hurts on Synthetic QA at 4 layers; overfitting gaps are negligible.

E Activation Function, Optimizer, and Loss Function Ablations

Figure 6 shows that ReLU, GELU, ELU, and LeakyReLU all achieve mean AUROC $\geq$0.998. Tanh is uniquely unstable during training on Synthetic QA, exhibiting persistent oscillations in validation AUROC throughout 140 epochs and achieving the lowest mean AUROC (0.994). Figure 7 shows that Adam and AdamW converge within $\sim$20 epochs to near-perfect AUROC, while SGD with momentum requires $\sim$60 epochs and RMSprop exhibits early volatility; all reach final AUROC $\geq$0.99. Figure 8 shows that label smoothing (0.05) provides a marginal positive gain over BCE on

Synthetic QA (+0.0008), while focal loss variants show small negative effects; HaluEval Sim and SciQ Sim are insensitive to loss function choice. The practical conclusion across all three ablations is that Neural DAC is robust to most design choices except Tanh activation and low learning rates.

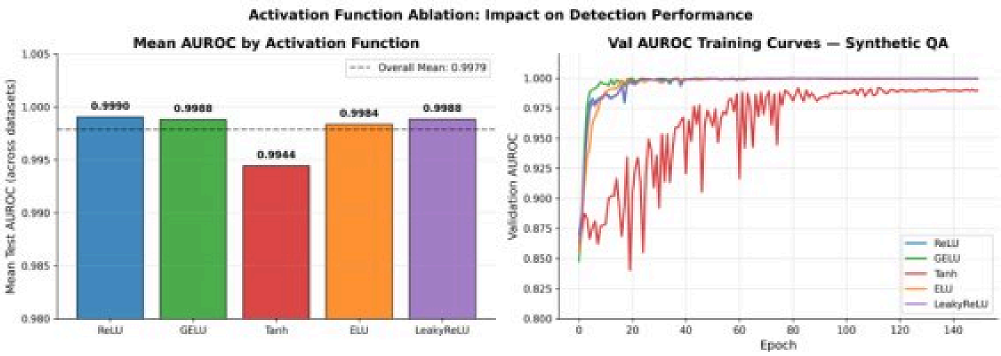


Figure 6: Activation function ablation: mean AUROC across datasets (left) and validation AUROC curves for Synthetic QA (right). Tanh is uniquely unstable with persistent oscillations; all other activations converge reliably to near-perfect AUROC ($\geq$0.998).

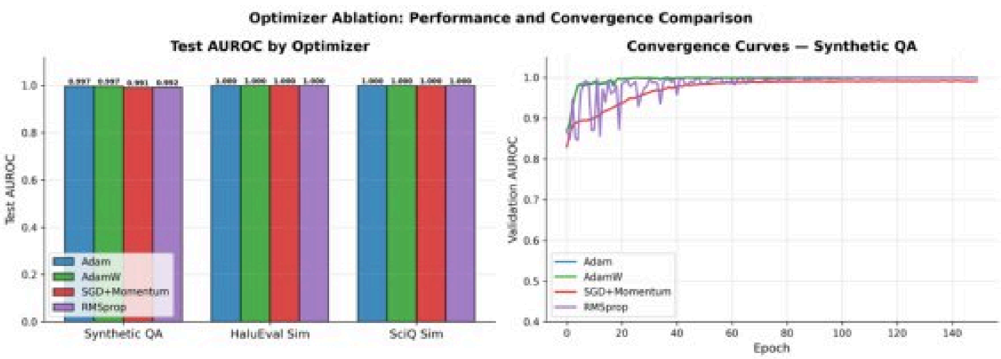


Figure 7: Optimizer convergence curves on Synthetic QA (right) and test AUROC across datasets (left). Adam and AdamW converge fastest ($\sim$20 epochs); SGD with momentum is slower ($\sim$60 epochs); RMSprop shows early volatility. All reach final AUROC $\geq$0.99 across all datasets.

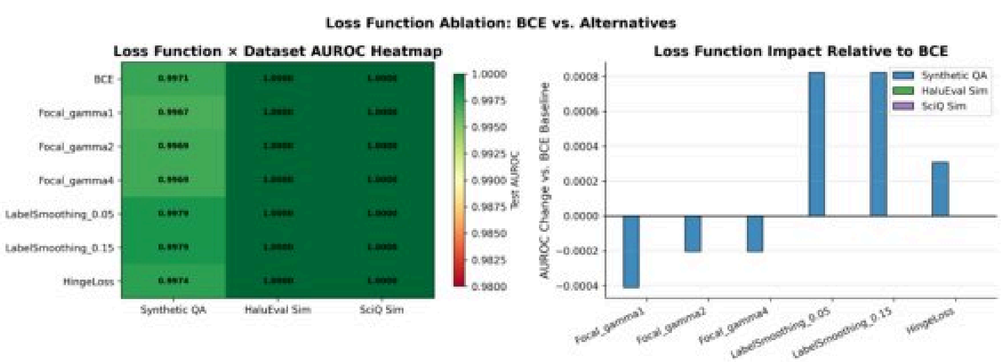


Figure 8: Loss function impact relative to BCE baseline (right) and absolute AUROC heatmap (left). Label smoothing (0.05) provides marginal gains on Synthetic QA (+0.0008); focal loss variants show small negative effects. All differences are $\leq$0.0008 AUROC, confirming that loss function choice is essentially irrelevant in this near-ceiling performance regime.

F Ensemble Methods

Figure 9 shows results for ensemble variants of Neural DAC (probability averaging, majority voting, and stacking logistic regression over multiple runs). Probability averaging and stacking provide essentially zero gain over the single model baseline. Notably, majority voting *degrades* performance on Synthetic QA by -0.93% relative to the single model—a negative result that highlights the risk of using hard-decision ensembling when individual model confidence is already near-perfect. Given this negative result alongside the negligible gains of other ensemble methods, ensembling is not recommended for this task.

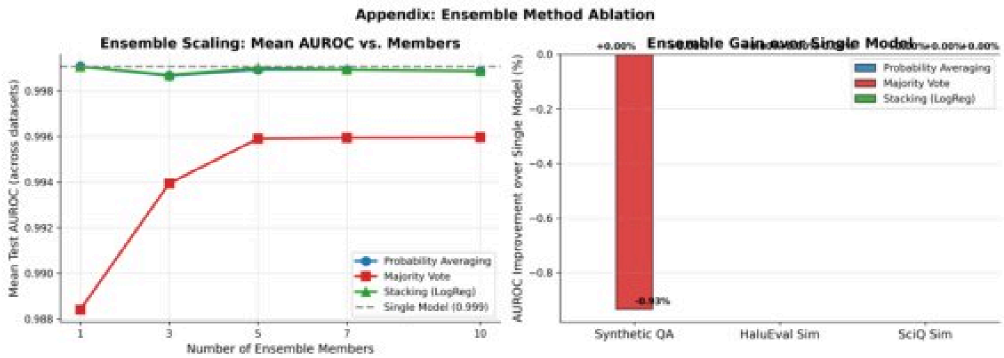


Figure 9: Ensemble method ablation: AUROC vs. number of ensemble members (left) and gain over single model (right). Probability averaging and stacking provide negligible gains; majority voting degrades performance on Synthetic QA by -0.93%, demonstrating that ensembling can hurt when individual models already achieve near-perfect discrimination.

G Data Efficiency

Figure 10 shows AUROC vs. fraction of training data used. DAC (LogReg) achieves near-ceiling AUROC from the very start on all datasets, confirming that drift features are highly informative and require minimal supervision. Neural DAC requires ∼25% of training data on Synthetic QA to reach plateau performance, but is near-ceiling from the start on HaluEval Sim and SciQ Sim. Both methods substantially outperform single-turn confidence regardless of training data fraction.

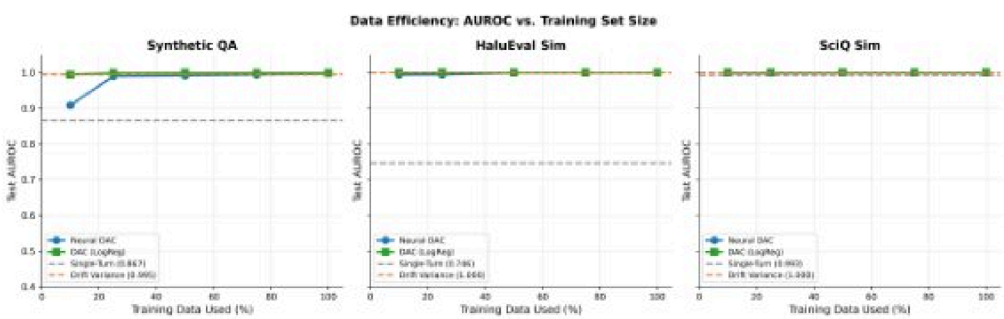


Figure 10: Data efficiency: AUROC vs. fraction of training data. DAC (LogReg) reaches near-ceiling performance with very little data on all datasets. Neural DAC requires slightly more data on Synthetic QA ($\sim$25%) but plateaus quickly. Both methods substantially outperform single-turn confidence at all data fractions.